

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ad5f571c6705e1a7e.jsonl`  
- SHA-256: `1ca68039ec7840a7c4cf8d39be214d8532a7386a9e992627ca601e2877220758`  
- Source modified: `2026-06-16T03:20:06+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:19:16.229Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://arxiv.org/html/2604.15173v1>

****Title:**** Boundary-Centric Active Learning for Temporal Action Segmentation (B-ACT, 2026)

****Found via:**** Small-data / weak / timestamp-supervised TAL + active learning search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:19:19.042Z)

I'll fetch the source and extract relevant claims.

3. user (2026-06-16T03:19:42.287Z)

B-ACT: Boundary-Centric Active Learning for Temporal Action Segmentation

1. Publication Date

****arXiv:2604.15173v1 [cs.CV] 16 Apr 2026****

2. Authors and Institution

3. Core Method

B-ACT operates through a two-stage hierarchical loop:

Stage 1 - Video Selection: The framework ranks unlabeled videos using predictive uncertainty via Monte Carlo Dropout. "Videos with high average uncertainty $U(j)$ contain more frames where the model is uncertain, suggesting higher potential gain from annotation."

Stage 2 - Boundary-Centric Clip Selection: Within selected videos, the method:

- Detects candidate action boundaries where predicted labels change
- Scores boundaries using a composite metric (Eq. 13) combining:
 - **Local uncertainty:** Average predictive entropy within a temporal window
 - **Confidence gap:** Margin between top-1 and top-2 class probabilities (inverted)
 - **Temporal gradient:** L2 distance between consecutive predicted distributions

Annotation Protocol: "Only the center-frame label $y_{bk(i)}(i)$ for each clip" is annotated, while surrounding unlabeled frames provide temporal context during training. Per-video cost = K labeled frames (typically $K=5$).

4. Specific Reported Numbers

Datasets: GTEA (28 videos, 11 classes), 50Salads (50 videos, 17 classes), Breakfast (1,712 videos, 48 classes)

Metrics: Frame-wise Accuracy, Segmental Edit Score, $F1@\{10, 25, 50\}$ (overlap thresholds)

Label Budgets:

- GTEA: 0.5% of training frames
- 50Salads: 0.16% of training frames
- Breakfast: 0.16% of training frames

Key Results (Table I):

Dataset	Method	$F1@50$	Edit	Accuracy
GTEA	Random	25.5	54.1	45.2
GTEA	Su et al. (SOTA baseline)	27.3	57.0	47.6
GTEA	B-ACT	42.2	66.6	61.5
50Salads	Su et al.	32.9	45.0	57.8
50Salads	B-ACT	52.6	56.7	73.2
Breakfast	Su et al.	43.5	58.6	63.5
Breakfast	B-ACT	50.9	67.5	70.5

B-ACT substantially outperforms prior work (Su et al.) with improvements of ~15 F1 points on GTEA and ~20 accuracy points on 50Salads.

5. Over-segmentation and Boundary Detection Claims

The paper emphasizes that "segmentation failures concentrate at ambiguous action transitions" and notes that "many TAS methods explicitly encode the prior that local temporal continuity should only change at true boundaries."

Regarding boundary-aware architectures, the authors state: "Boundaries are more genuine?and more critically in need of supervision?if the model's predicted distribution shifts sharply across it," addressing the over-segmentation failure mode through temporal gradient scoring.

6. Data Efficiency Claims

B-ACT demonstrates extreme label efficiency:

- GTEA: Only **0.5%** of frames labeled (30% initialization + 4 rounds $\times$ 25% querying)

- 50Salads: Only ****0.16% of frames**** labeled
- Breakfast: Only ****0.16% of frames**** labeled

"Each queried video is annotated with single-frame clips ($\tau=1$), corresponding to $\tau 0.4\%$ of frames per video" for GTEA and " $\tau 0.13\%$ of frames per video" for 50Salads.

Despite sparse supervision, B-ACT achieves strong performance gaps: on 50Salads at 0.16% budget, it reaches 73.2% accuracy versus 57.8% for the previous best method.

7. Related Supervision Paradigms

****Timestamp Supervision:**** The paper notes that "timestamp-based supervision further reduces labeling density by training from a small set of annotated time points per action" and claims their approach is "strictly more informative than timestamp supervision: we also supply τ unlabeled context frames around each boundary."

****Semi-supervised Learning:**** "Prior studies on TAS have investigated semi-supervised approaches that leverage a limited set of fully annotated videos alongside a larger collection of unlabeled videos" but "these methods still depend on fully annotating multiple untrimmed videos."

****Weak Supervision:**** The authors mention weakly-supervised methods rely on "coarse annotations for all training videos" and notes their "performance of both semi-supervised and weakly-supervised approaches are significantly lower than the fully supervised methods."

4. assistant (2026-06-16T03:19:46.896Z)

The fetch succeeded with detailed, specific content. This is a primary arXiv research paper directly relevant to small-data TAL with active learning, boundary detection, and over-segmentation control τ all four research dimensions. Note: the arXiv ID (2604.xxxxx = April 2026) and date are slightly future-dated relative to my knowledge cutoff but consistent with the session date (2026-06-16); I'll report the date as the source states it.

5. user (2026-06-16T03:20:00.507Z)

Structured output provided successfully

6. assistant (2026-06-16T03:20:06.517Z)

Extracted and returned. Source is a primary arXiv research paper (B-ACT, arXiv:2604.15173v1, dated 16 Apr 2026 per the page) directly on-target for dimensions 1 (boundary detection/over-segmentation), 3 (small-data active learning + timestamp/weak supervision), and 4 (segmental F1@k / edit-score evaluation).

One verification caveat I flagged in the output: the arXiv ID (2604.xxxxx) and the stated date are future-relative to my training cutoff but consistent with the session's current date. The reported numbers are quoted directly from the fetched page; I could not independently cross-check the Table I figures against the PDF. The institution for the authors (Halil Ismail Helvacı, Sen-ching Samson Cheung) was not present in the fetched content.